

Towards a self-cognitive complex product design system: A fine-grained multi-modal feature recognition and semantic understanding approach using large language models in mechanical engineering

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ABSTRACT

Facing the promising tendency of human-artificial intelligence (AI) collaborative product design, fine-grained and multi-modal mechanical part recognition and semantic understanding have become a basic task for achieving a self-cognitive product design system. However, traditional semantic understanding approaches for mechanical parts can only handle single-modal data, which is either textual or image data, resulting in the following limitations 1) insufficient mining on fine-grained part's functional/behavioral/structural information, and 2) ineffectiveness on multi-modal part information alignment, therefore restricting the intelligence level of the previous product design assistants. To mitigate these challenges, this paper proposes a fine-grained multimodal reasoning approach for mechanical part semantic understanding. The proposed approach utilizes a pre-trained Convolutional Neural Network (CNN) for visual feature extraction, a large language model (LLM) called LLaMA3 for advanced textual analysis, and a Unified Feature Fusion Module (UFFM) to facilitate robust cross-modal interactions. A positive and negative sample generation mechanism is implemented to refine the model's ability to discern subtle variations in complex components. Experimental evaluations on the Industrial Part Multimodal Dataset (IPMD) demonstrate a significant improvement in classification accuracy, providing a more precise and intelligent solution for the semantic understanding in complex product design systems.

1. Introduction

Facing the promising tendency of human-artificial intelligence (AI) collaborative product design, the semantic recognition and comprehension of mechanical parts are crucial for achieving the cognitive capability of intelligent design systems [1,2]. Mechanical parts, as the basic building blocks of products, their fine-grained multi-modal recognition and semantic understanding are the cornerstones of constructing an advanced design system. However, the structure and function of mechanical parts themselves are becoming increasingly complex, resulting in the property associations between parts becoming more intricate and difficult to distinguish [3]. Which exposes many disadvantages of traditional methods in the key task of semantic understanding of machine parts. First, traditional methods usually can only handle single-modal data, making it difficult to integrate multi-modal information for comprehensive analysis and unable to fully utilize the complementary advantages of different modal data [4]. Second, traditional methods relying on predefined rules and fixed query patterns are inefficient in mining the fine-grained functional, behavioral, and structural information of parts [5]. When faced with similar parts in the same product

series, it is difficult to accurately distinguish their detailed structural differences, resulting in insufficient fine-grained understanding of parts. Third, traditional methods lack the ability to align multi-modal part information. This leads to insufficient semantic understanding, making the model more sensitive to noise and thus reducing the recognition accuracy [6]. Large Language Models (LLMs) are pre-trained on large-scale corpora. Compared with traditional language processing methods, they possess more powerful semantic representation and generation capabilities [7]. However, due to the insufficient adaptability of the pre-trained data to the field of mechanical engineering and the lack of in-depth understanding of professional knowledge such as manufacturing processes, material properties, and complex assembly relationships, LLMs have limitations when dealing with professional problems [8,9].

To address these issues, considering the limitation of single-modal processing in traditional methods, this paper proposes to use a pre-trained Convolutional Neural Network (CNN) to extract visual features from mechanical part images. At the same time, the LLaMA3 model is used for advanced text analysis, and a Unified Feature Fusion Module

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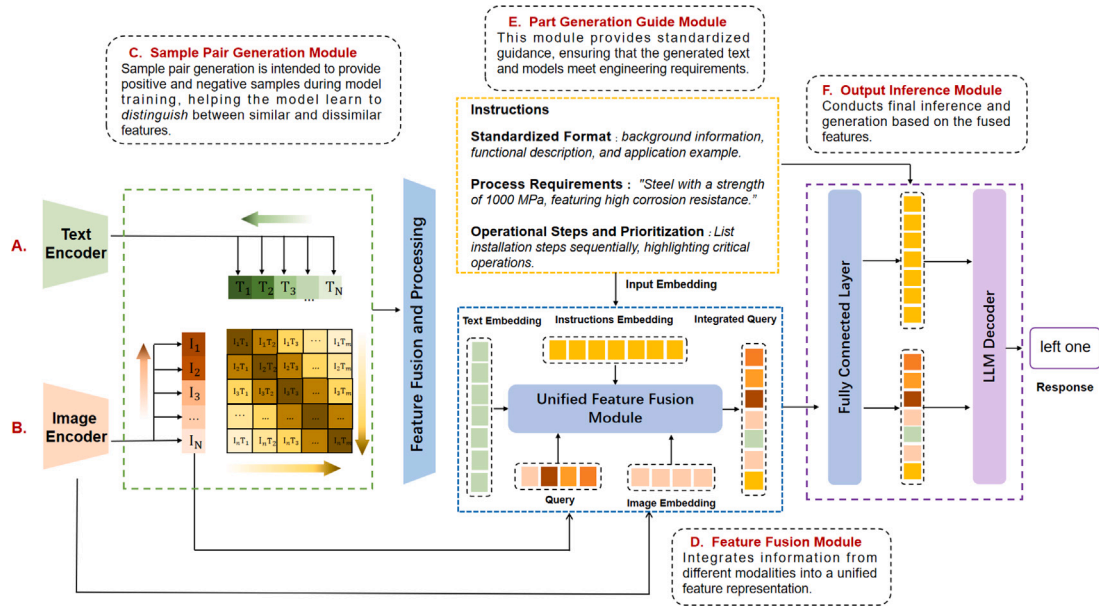


Fig. 1. Overview of the proposed multimodal fine-grained reasoning model architecture.

(UFFM) is introduced to effectively fuse visual and text modal information, breaking through the bottleneck of single-modal processing. To solve the problem of difficult fine-grained information mining, a positive-negative sample generation mechanism is developed. Through a carefully designed sample generation strategy, the model's ability to distinguish subtle differences in complex components is enhanced, enabling it to more effectively mine the fine-grained functional, behavioral, and structural information of parts. In solving the problem of insufficient multi-modal information alignment, the self-attention and cross-attention mechanisms in UFFM are used to enhance cross-modal information alignment, providing high-quality input for subsequent reasoning modules, thereby improving the accuracy of semantic understanding, reducing the model's sensitivity to noise, and increasing the recognition accuracy.

The contributions of this study are mainly reflected in the following aspects: 1. A comprehensive multi-modal feature recognition framework is constructed, which combines CNN, LLMs, and UFFM, effectively overcoming the limitations of traditional single-modal methods. 2. UFFM improves the alignment of cross-modal information through self-attention and cross-attention mechanisms, providing high-quality input for subsequent reasoning modules. 3. An Industrial Part Multi-Modal Dataset (IPMD) is constructed, which covers multi-modal information of various industrial part types, providing rich and high-quality data support for research related to semantic understanding of mechanical parts.

The subsequent content of this paper is arranged as follows: Section 2 reviews relevant research work; Section 3 elaborates on the proposed method in detail; Section 4 presents the experimental settings and results; Section 5 conducts a case study; finally, Section 6 summarizes the research results and explores future research directions.

2. Relate work

2.1. Multimodal fusion in mechanical engineering

Multimodal fusion technology is crucial for the semantic understanding of mechanical components. In the early days, multimodal fusion mainly involved simply concatenating data from different modalities into feature vectors [10]. However, due to the significant differences in features and distributions of data from different modalities,

this approach was prone to information loss and conflicts, and had low efficiency in processing large-scale data [11]. With the rise of deep learning, multimodal models represented by CLIP fuse image and text features through contrastive learning, and can complete various cross-modal tasks without fine-tuning [12]. BLIP proposed a unified framework to achieve image text retrieval and generation [13]; BLIP2 is further optimized on this basis [14]. However, when these general-purpose architectures are applied to heterogeneous data such as CAD models and maintenance logs in the engineering field, standard fusion strategies such as feature concatenation and cross-attention are difficult to apply. Although some studies have attempted domain-specific pre-training to address this, the problems of multimodal data alignment and fusion in industrial environments remain unsolved [15]. Coupled with the difficulties in obtaining multimodal data in the mechanical field, high annotation costs, and spatio-temporal inconsistencies, the challenges of data alignment and synchronization are enormous [16].

2.2. Fine-grained recognition from general images to mechanical components

Fine-grained recognition is the key to achieving high-precision semantic understanding of mechanical components. In the early days, it relied on hand-crafted features and traditional machine learning algorithms [17]. Due to the high dimensionality of features, data sparsity, and low extraction efficiency, these methods had insufficient descriptive ability for complex features [18]. With the development of deep learning, CNN-based methods automatically learn multi-level features through multi-layer convolution and pooling [19]. The feature pyramid structure can capture subtle features at different scales, improving the ability to distinguish similar features [20]. The attention mechanism enables the model to focus more on key regions and optimize the extraction of features with subtle differences [21]. Nowadays, advanced methods such as TransFG [22] and ProtoCon [23] use the attention mechanism and contrastive learning to distinguish subtle category differences and perform well on biological datasets. However, their performance deteriorates when applied to mechanical components with overlapping geometric shapes. The main problems are over-reliance on visual saliency, neglect of text metadata, and poor negative sampling effects, making it difficult to generate effective samples for similar parts.

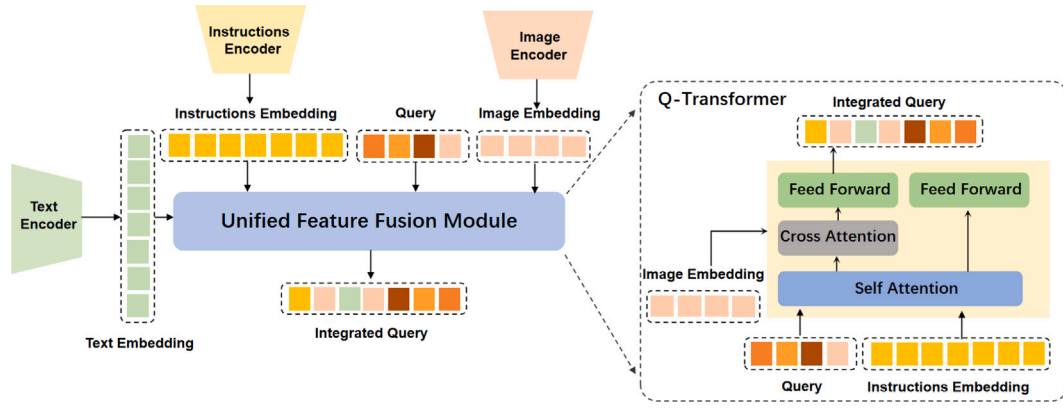


Fig. 2. Feature fusion and processing mechanism using UFFM.

2.3. Semantic understanding in machinery

Semantic understanding is the ultimate goal of research on mechanical component semantic understanding. In the early days, based on text classification and image semantic annotation techniques, the understanding of semantic relationships was relatively shallow, and it was difficult to handle complex semantics [24]. Models based on the Transformer architecture capture semantic relationships through the multi-head self-attention mechanism [25]. Combining the attention mechanism with knowledge graphs enhances the ability to understand complex semantics [26,27]. LLMs such as GPT-4 and LLaMA have brought new directions for the analysis and semantic understanding of technical documents [28]. The latest research trend, inspired by CLIP-based architectures, attempts to bridge the gap between text and vision, but has limited scalability when dealing with domain-specific terms [29]. Although LLMs have strong general-purpose language understanding capabilities, their application in engineering requires fine-tuning to adapt to the precision and structured characteristics of technical documents [30]. Currently, how to effectively integrate LLMs with multimodal data for mechanical part recognition remains an issue to be studied [31].

3. Methodology

3.1. Overview of the methodology

This study introduces a novel multimodal fine-grained reasoning approach designed to enhance the semantic understanding of mechanical components in complex industrial settings. The proposed model architecture, illustrated in Fig. 1, comprises six integral modules: the Text Encoder, Image Encoder, Sample Pair Generation Module, Feature Fusion Module, Part Generation Guide Module, and Output Inference Module. The specific functionalities of each module are described as follows:

A. Text Encoder: This module employs the LLaMA3 model to encode textual descriptions of mechanical parts into high-dimensional feature vectors. The multi-head attention mechanism is employed to conduct parallel processing of the technical documents related to mechanical parts. It demonstrates a robust language understanding capability, enabling the recognition of the fundamental properties of parts and the mastery of intricate assembly and operation processes. Consequently, the complex semantic dependencies and context information can be comprehensively and deeply understood. High-quality text features are furnished for subsequent multimodal fusion.

B. Image Encoder: The ResNet model is adopted. The problem of gradient vanishing during training is addressed through the residual connection. This model enables the comprehensive capture of visual features such as the overall contour, details, and surface texture of parts

from deep convolutional layers, providing comprehensive and accurate visual information for subsequent feature fusion.

C. Sample Pair Generation Module: This module generates sample pairs through hard negative mining and cross-modal contrastive learning techniques. During hard negative mining, negative samples highly similar to positive samples are selected in the feature space. This selection enables the model to learn the subtle differences between samples. Cross-modal contrastive learning, on the other hand, explores cross-modal semantic associations by matching features between the text and image modalities.

D. Feature Fusion Module: As a core component for multi-modal information interaction in the model, the UFFM is utilized to deeply integrate textual, visual, and instructional embeddings. By leveraging the self-attention mechanism, it deeply combs through the features within each modality, uncovering long-range dependencies within the modality. The cross-attention mechanism, meanwhile, enables the interaction and alignment of features across different modalities.

E. Part Generation Guide Module: This module ensures that the generated outputs adhere to established standards within the engineering domain. By incorporating standardized formats, process specifications, and operational guidelines, the Part Generation Guide Module guarantees that the model's outputs are not only theoretically sound but also practical and applicable to real-world engineering contexts, thereby bridging the gap between model outputs and industry requirements.

F. Output Inference Module: Tasked with advanced reasoning and content generation based on the fused multimodal features, this module employs fully connected neural layers and a LLM decoder to produce high-quality outputs tailored to specific task demands. The module's architecture allows it to generate results that are highly contextualized and aligned with the particular requirements of complex industrial applications.

3.2. Feature fusion and processing

The UFFM as illustrated in Fig. 2. First, the text encoder and image encoder respectively extract the feature embeddings of the descriptive text and image, represented as text embeddings f_T and image embeddings f_I , which are then fed as the initial input into the UFFM. Within the UFFM, the self-attention mechanism is employed for deep feature extraction within each modality. The calculation formula is as follows:

$$\text{Self-Attention}(f_x) = \text{softmax}\left(\frac{f_x f_x^T}{\sqrt{d_x}}\right) f_x \quad (1)$$

Where f_x represents the input features, and d_x is the dimensionality of the feature vectors. The self-attention mechanism ensures that the model deeply understands the independent features of each modality. Subsequently, the cross-attention mechanism promotes the interactive

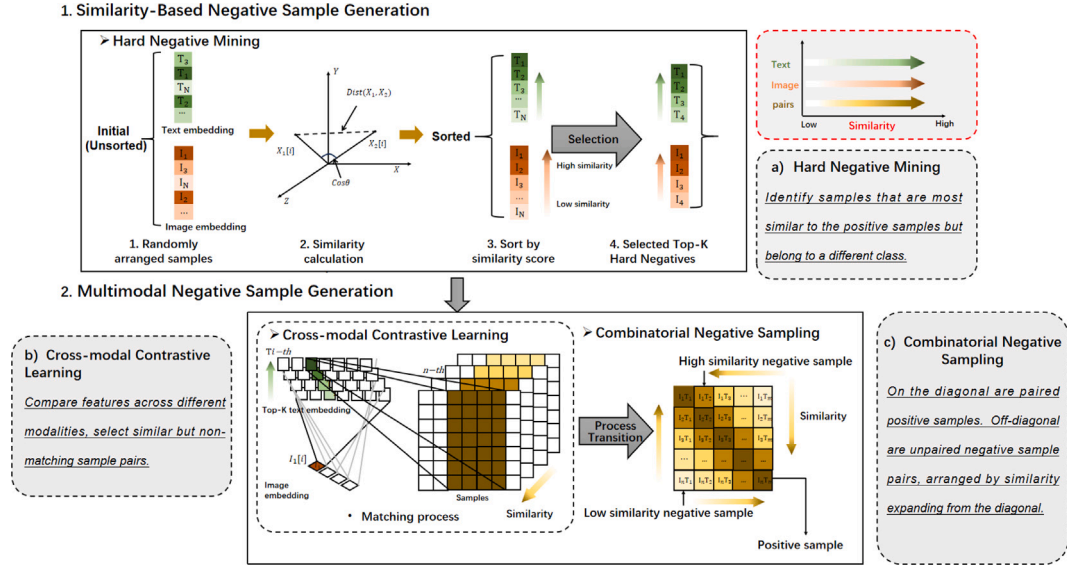


Fig. 3. Multimodal sample pair generation mechanism.

learning of text and image features, allowing the model to analyze and interpret the complex relationships between different modalities.

The inter-attention mechanism measures the degree of correlation between the features of text and image by calculating the attention score matrix *AttentionScore*. The calculation formula is:

$$AttentionScore_{ij} = \frac{\exp(f_{T,i}^T f_{I,j})}{\sum_{k=1}^n \exp(f_{T,i}^T f_{I,k})} \quad (2)$$

where $f_{T,i}$ represents the i th feature vector in the text embedding f_T , $f_{I,j}$ represents the j th feature vector in the image embedding f_I , and n represents the number of feature vectors in the image embedding f_I . This score reflects the correlation between text and image features, and the higher the score, the stronger the correlation.

Based on this attention score matrix, the model performs a weighted sum on the image embedding f_I to obtain the text-related image feature representation $f_{I,T-related}$. The calculation formula is as follows:

$$f_{I,T-related} = \sum_{j=1}^n AttentionScore_{ij} f_{I,j} \quad (3)$$

The resulting fused multimodal feature representation is formulated as follows:

$$f_{fusion} = Cross-Attention(f_T, f_I) \quad (4)$$

The fused feature f_{fusion} captures the deep semantic relationships of multimodal information. Through the aforementioned mechanisms, UFFM enhances the alignment and synergy between modalities, ensuring that the generated multimodal features provide high-quality input for subsequent inference modules.

3.3. Sample pair generation mechanism

As illustrated in Fig. 3, this study proposes a multimodal sample generation mechanism to enhance the model's fine-grained reasoning ability in complex industrial scenarios. This mechanism consists of two main components: similarity-based negative sample generation and multimodal negative sample generation.

Hard Negative Mining: First, similarity calculations are performed separately within each single modality. As shown Fig. 3(a), the model processes the initial state of text embeddings f_T and image embeddings f_I independently, constructing their respective feature spaces through similarity computations. Similarity be measured using the Euclidean distance formula:

3.3.1. Similarity-based negative sample generation

$$Sim(f_a, f_b) = \|f_a - f_b\|_2 \quad (5)$$

Within the feature space, the model ranks negative samples f_T^- based on their distance from the positive sample f_T^+ . A threshold K is set, and the top K samples that are most similar to the positive sample are selected as hard negative samples. This process ensures that the model can effectively distinguish subtle feature differences in single-modality tasks, thereby enhancing fine-grained recognition capabilities.

$$Hard\ Negatives = \{f_T^- \mid Sim(f_T^+, f_T^-) \leq \epsilon \text{ for top } K\} \quad (6)$$

ϵ is the similarity threshold used to filter out the closest negative samples.

3.3.2. Multimodal negative sample generation

Cross-modal Contrastive Learning: In the cross-modal contrastive learning phase, the model initially performs feature matching between text embeddings f_T and image embeddings f_I . Specifically, as shown in Fig. 3(b), the model filters out the top K text embeddings most similar to the image embeddings based on similarity scores. The selected text embeddings represent the textual descriptions most closely aligned with the target image features in the feature space. The similarity scoring formula is:

$$Sim_{cross}(f_{T,i}, f_{I,j}) = \frac{f_{T,i} \cdot f_{I,j}}{\|f_{T,i}\| \|f_{I,j}\|} \quad (7)$$

Subsequently, the model conducts fine-grained feature comparisons between the Top K text embeddings and image embeddings across n th dimensions to capture deeper cross-modal similarities. By utilizing a layer-by-layer matching approach, the model effectively identifies negative sample pairs that appear similar on the surface but are semantically mismatched. The optimized contrastive loss function is as follows:

$$\mathcal{L}_{contrastive} = \sum_{i,j} \left[y_{ij} \|f_{T,i} - f_{I,j}\|_2^2 + (1 - y_{ij}) \max(0, m - \|f_{T,i} - f_{I,j}\|_2)^2 \right] \quad (8)$$

In this context, y_{ij} is an indicator function that determines whether the sample pair (i, j) is a positive sample pair, and m represents the margin parameter for contrastive learning.

Table 1
Multimodal data fusion instructions for vacuum sucker.

Vacuum sucker	
Feature selection	1. Prioritize the outer diameter and thickness of the vacuum sucker bellows for optimal suction force. 2. Emphasize the material composition for durability and adaptability in high-temperature environments. 3. Retain structural descriptions that impact performance, such as the sealing ring and attachment points.
Weight distribution	1. Assign 50% weight to geometric parameters from engineering drawings due to their critical role in ensuring fit and function. 2. Distribute the remaining weight equally between 3D images (20%) and physical images (30%) to balance shape verification and material property validation.
Feature alignment	1. Align suction cup diameter from engineering drawings with measurements in 3D models. 2. Match surface finish details from physical images with material properties mentioned in textual descriptions. 3. Ensure the alignment of mounting holes and structural integrity as depicted in 3D images with real-world usage requirements. 4. Validate that the sealing effectiveness depicted in the 3D images matches the required industrial standards.

Table 2
Multimodal data fusion instructions for guide shaft.

Guide Shaft	
Feature selection	1. Focus on the shaft diameter and length, as these are crucial for fitting into assembly configurations. 2. Highlight the surface coating type to determine resistance to wear and environmental factors. 3. Retain specific threading details which are essential for proper assembly with other mechanical parts. 4. Include bearing seat dimensions to ensure compatibility with various bearing types. 5. Capture material hardness specifications that affect load-bearing capacity.
Weight distribution	1. Assign 40% weight to dimensional data from engineering drawings, as these are key to precision. 2. Allocate 30% weight to 3D models, focusing on visual structural validation. 3. Assign 30% to physical images to confirm material and surface treatment details.
Feature alignment	1. Align key shaft diameters from engineering drawings with dimensions observed in 3D renderings. 2. Match surface treatment and coating features between physical images and textual descriptions. 3. Verify alignment of threading specifications across text, drawings, and 3D models. 4. Cross-check bearing seat dimensions against assembly requirements in the textual guidelines. 5. Align tolerance specifications across all modalities to ensure consistency in manufacturing. 6. Confirm that the mechanical fit observed in 3D models aligns with real-world usage scenarios.

Table 3
Multimodal data fusion instructions for universal ball.

Universal Ball	
Feature selection	1. Select key parameters such as ball diameter and pivot axis position for accurate motion analysis. 2. Retain details of surface finish, crucial for ensuring low friction and wear resistance. 3. Include mounting feature dimensions to validate ease of assembly in different configurations. 4. Highlight the material type to confirm load-bearing capabilities under different operational stresses.
Weight distribution	1. Assign 45% weight to geometric parameters from engineering drawings due to their critical importance. 2. Distribute remaining weight evenly between 3D images (25%) and physical images (30%) for balanced feature validation.
Feature alignment	1. Match the key ball joint diameters from engineering drawings with 3D models. 2. Verify the consistency of material properties across physical images and textual descriptions. 3. Align mounting hole positions across modalities to ensure compatibility with various configurations. 4. Cross-reference surface treatment data with functional descriptions to ensure durability. 5. Confirm that real-world images demonstrate the expected mechanical behavior under stress conditions as depicted in the engineering data.

Combinatorial Negative Sampling: During the combinatorial negative sampling stage, the model generates complex negative sample pairs by combining image embeddings f_I and text embeddings f_T from different modalities, as illustrated in Fig. 3(c). These combined sample pairs are sorted by similarity from low to high. The model prioritizes learning from highly similar negative sample pairs to enhance its ability to discern fine-grained differences. The generation formula for the combined samples is as follows:

$$\text{Sim}(f_{I,i}, f_{T,j}) = \|f_{I,i} - f_{T,j}\|_2 \quad (9)$$

In this formula $f_{T,i}$, denotes the i th feature vector within the image embedding f_I , and $f_{T,j}$ represents the j th feature vector in the text embedding f_T . The Euclidean distance calculated here serves as a measure of similarity. A smaller distance indicates a higher degree of similarity between the two feature vectors, signifying that the corresponding sample pair is more alike.

By combining multimodal features, the model generates more challenging negative sample pairs, enhancing robustness and performance in multimodal semantic understanding.

3.4. Part instruction generation module

The Part Instruction Generation Module plays a pivotal role in the multimodal fine-grained reasoning model, translating complex technical specifications into actionable guidance documents tailored for practical engineering applications. This module operates during both the multimodal data fusion and output inference stages.

3.4.1. Multimodal data fusion stage

During the multimodal data fusion stage, the Part Instruction Generation Module ensures the coordinated handling of text, image, and query vectors through feature selection instructions, priority and weight assignment instructions, and feature alignment instructions to meet practical engineering needs. The multi-modal data fusion instructions specifically designed for the three parts of vacuum Sucker, guide shaft and universal ball are shown in Tables 1–5.

Feature Selection Instructions: The module prioritizes essential features for engineering tasks through feature selection strategies. For engineering drawings, key dimensional parameters critical to manufacturing (e.g., suction cup diameter, threaded hole distribution) are

Table 4
Multimodal data fusion instructions for slide rail.

Slide rail	
Feature selection	1. Select key parameters like rail length, cross-sectional dimensions, and slider groove shape for accurate motion and load-bearing analysis. 2. Retain details of surface hardness and roughness, crucial for smooth sliding and wear resistance. 3. Include mounting hole dimensions and spacing to validate assembly in different set-ups. 4. Highlight the material type (e.g., steel, aluminum alloy) to confirm load-bearing and corrosion-resistance capabilities.
Weight distribution	1. Assign 40% weight to geometric parameters from engineering drawings due to their critical importance in defining rail functionality. 2. Distribute the remaining weight evenly between 3D images (30%) and physical images (30%) for balanced feature validation.
Feature alignment	1. Match the key rail dimensions and slider interface features from engineering drawings with 3D models. 2. Verify the consistency of material properties across physical images and textual descriptions. 3. Align mounting hole positions across modalities to ensure compatibility with different mechanical configurations. 4. Cross-reference surface treatment data (e.g., coating type) with functional descriptions to ensure durability and smooth operation. 5. Confirm that real-world images demonstrate the expected sliding behavior and load-bearing capacity as depicted in the engineering data.

Table 5
Multimodal data fusion instructions for valve.

Valve	
Feature selection	1. Select key parameters such as valve type (e.g., gate, globe, check), nominal diameter, and pressure-rating for accurate flow control and system compatibility analysis. 2. Retain details of seal material and quality, crucial for preventing leakage. 3. Include connection dimensions (e.g., flange size, thread type) to validate installation in different pipelines. 4. Highlight the body material type to confirm pressure-resistance and corrosion-resistance capabilities under different fluid conditions.
Weight distribution	1. Assign 45% weight to geometric and specification parameters from engineering drawings as they are fundamental for valve operation. 2. Distribute remaining weight evenly between 3D images (25%) and physical images (30%) for balanced feature validation.
Feature alignment	1. Match the key valve dimensions and internal structure features from engineering drawings with 3D models. 2. Verify the consistency of material properties and seal characteristics across physical images and textual descriptions. 3. Align connection dimensions across modalities to ensure proper fit in pipeline systems. 4. Cross-reference surface treatment and coating data with functional descriptions to ensure long-term reliability and resistance to fluid erosion. 5. Confirm that real-world images demonstrate the expected flow-control behavior and pressure-handling capacity as depicted in the engineering data.

retained; for 3D images, the overall structural descriptions are preserved to verify shape features; and for real-world images, focus is placed on material properties and surface treatments.

Weight Assignment Instructions: Based on the engineering value of different modal data, the module assigns corresponding weights and priorities. In this model, engineering drawings, which provide precise geometric parameters and are fewer in number, are given the highest weight of 50%; 3D images, offering overall structural information and being the most abundant, are assigned a 20% weight; and real-world images, used to confirm material and surface properties and available in moderate numbers, receive a 30% weight.

Feature Alignment Instructions: During multimodal data fusion, these instructions match key information from text descriptions with corresponding features in images. This includes aligning dimensional parameters from text with geometric data in engineering drawings, matching shape descriptions with 3D structures, and correlating material descriptions with actual material displays in real-world images.

3.4.2. Output inference stage

In the output inference stage, the module ensures that the multimodal features are precisely translated into engineering-compliant output documents. This is achieved through parameterized instructions, standardized formatting, and scenario adaptation to guide the generation of high-value engineering text, and the output instructions for various types of parts are shown in Table 6.

Standardized Format: The module rigorously controls content norms and formatting to ensure consistency and professionalism in technical documentation. It specifies formats for titles, paragraphs, and lists, standardizes language style, employs commonly used engineering terms, and avoids subjective descriptions, ensuring the document's readability and applicability.

Operational Guidance: The module analyzes part parameters and generates detailed operational guidance that meets process requirements, including installation and operational steps. Each step not only provides basic installation instructions but also emphasizes the priority of critical procedures and key precautions, ensuring compliance with engineering standards.

Step-by-Step Process Instructions: Using parameterized instructions, the module generates adaptable operational steps. It dynamically adjusts instructions according to specific part specifications such as diameter, material, hardness, and operating temperature.

Environment and Application Scenarios: The module generates flexible operational steps based on parameterized instructions, adapting the process instructions dynamically according to variations in part specifications like diameter, material, hardness, and working temperature, ensuring the outputs are well-suited to diverse engineering environments and usage scenarios.

4. Experiment

4.1. Experimental setup

4.1.1. Dataset

This study constructed a multimodal dataset, IPMD, specifically designed for the recognition and analysis of complex industrial parts. The dataset compilation involved the integration of image and text information.

Image Data: The images were primarily sourced from publicly available industrial part databases, on-site data collection in manufacturing plants, and some open-source CAD model repositories. After a rigorous screening process, they were classified based on part complexity and application scenarios. The dataset covered five categories of industrial parts: vacuum suckers, guide shafts, universal balls, slide rails and valves. The dataset, detailed in Table 7. There are six types of vacuum suckers, fourteen types of guide shafts, nine types of universal balls, fifteen types of slide rails, and three types of valves. In total, the dataset contains 2355 images. It encompasses three types of images: 3D models, real-world images, and engineering drawings. 3D models are useful for capturing the overall shape of the parts, engineering drawings provide precise dimensions and machining details, and real-world images are beneficial for demonstrating material properties and surface treatments. Each image in the dataset is meticulously annotated. The annotations include part names, key feature dimensions, and relevant manufacturing process information.

Table 6
Output inference stage instructions for different part types.

Part type	Instruction type	Guidance instructions
Vacuum sucker	Standardized Format	Generate content in a standardized format: include sections for background, function, and application. Use precise technical terms without subjective descriptions.
	Operational Guidance	Create operational steps for vacuum sucker installation: detail material specifications (e.g., rubber types), mounting instructions, and maintenance guidelines.
	Step-by-Step Process	Outline step-by-step instructions for assembling the vacuum sucker, focusing on key procedures like substrate securing, bolt tightening, and suction calibration.
	Environment Scenarios	Adjust steps for various environments such as high-temperature operations or uneven surfaces. Include operating temperature ranges and specific adaptation guidelines.
Guide shaft	Standardized Format	Format the technical guide consistently, using bullet points for clear steps and engineering terminology relevant to alignment and rotational components.
	Operational Guidance	Develop installation and operational instructions for guide shafts: include shaft alignment, lubrication requirements, and rotational stability measures.
	Step-by-Step Process	Create detailed assembly steps, highlighting the positioning, fitting tolerances, and alignment procedures for accurate shaft operation.
	Environment Scenarios	Provide adaptable guidance for use in environments with variable lubrication conditions or heavy-duty rotational requirements. Adjust based on shaft material properties.
Universal ball	Standardized Format	Maintain consistency in formatting, focusing on sections relevant to ball joint operation, including background, functional range, and assembly details.
	Operational Guidance	Generate detailed operational instructions for universal balls, including rotational joint assembly, lubrication tips, and surface treatment requirements.
	Step-by-Step Process	List sequential assembly steps for universal balls, emphasizing fitting accuracy, socket interface preparation, and rotational freedom adjustments.
	Environment Scenarios	Include adaptive instructions for different operational scenarios like high-friction or low-temperature environments. Outline steps to enhance performance consistency.
Slide rail	Standardized Format	Present rail type, load-bearing, and dimensions in a structured, bullet-pointed way.
	Operational Guidance	Cover slide rail pre-install, mounting, slider attach, and issue-handling.
	Step-by-Step Process	Sequence slide rail assembly steps, stressing positioning, fastening, and alignment.
	Environment Scenarios	Advise on slide rail use in dust, humidity, vibration, and offer harsh-condition solutions.
Valve	Standardized Format	Structure valve type, pressure, and connection info with clear headings.
	Operational Guidance	Detail valve selection, mounting, startup, flow/pressure regulation, and malfunction-handling.
	Step-by-Step Process	List valve assembly/disassembly steps, highlighting safety and torque specs.
	Environment Scenarios	Adapt guidance for high-pressure/temperature, corrosive env., suggest materials and schedules.

Data Augmentation: To enhance the diversity of the image data and improve the model's generalization ability, various data augmentation techniques were applied: random rotation (-15° to $+15^\circ$), random cropping (85%–100% area), color perturbations (brightness $\pm 20\%$, contrast $\pm 15\%$, saturation $\pm 15\%$, hue $\pm 10\%$), horizontal flipping (50% probability), scaling (0.8–1.2 ratio), Gaussian noise (standard deviation 0.01), and Gaussian blur (kernel size 3×3 , standard deviation 0.1–0.5). These augmentation strategies expanded the diversity of the data while preserving the core features of the parts, enhancing the model's robustness under varying viewpoints, lighting conditions, and other real-world scenarios.

Text Data: Text data were divided into two categories: descriptive text and instruction text. Descriptive text was used to describe image information, generated through a process shown in Fig. 4, including prompt design, text generation, and manual correction. Initially, specialized prompts were crafted based on image types; then, GPT-4 was used to generate preliminary text; this was followed by multiple rounds of expert verification and optimization to produce engineering-standard descriptive text. Instruction texts were further divided into multimodal fusion instructions and inference instructions. As summarized in Table 8. Text Data Composition and Instruction Types in IPMD, the dataset contains a total of 223 text entries.

4.1.2. Sample pair generation

Based on the constructed dataset, this study further generated positive and negative sample pairs to enhance the model's ability to

Table 7
Composition and distribution of images in IPMD.

Type	Categories	Number
Slide rails	15	750
Guide shaft	14	421
Universal ball	9	261
Vacuum sucker	6	771
Valves	3	152
Total	47	2255

discern fine-grained differences. Positive pairs consisted of matched multimodal text and image data representing the same part. Negative pairs were generated through hard negative mining and cross-modal contrastive learning, encompassing combinations of similar but different classes to strengthen the model's ability to learn subtle feature differences. Experimental results indicated that a positive-to-negative sample pair ratio of 1:3 achieved optimal training performance, with specific data outlined in Table 9.

Cross-Validation Strategy: Given the limited size and imbalanced class distribution of the dataset, a straightforward train-test split could lead to overfitting and biased performance evaluation, failing to accurately reflect the model's true capabilities. To enhance the generalizability and robustness of the results, this study employs a 5-fold cross-validation approach. Specifically, the dataset is partitioned into five non-overlapping subsets, or folds. In each iteration, one fold serves as the validation set while the remaining four folds are used for training

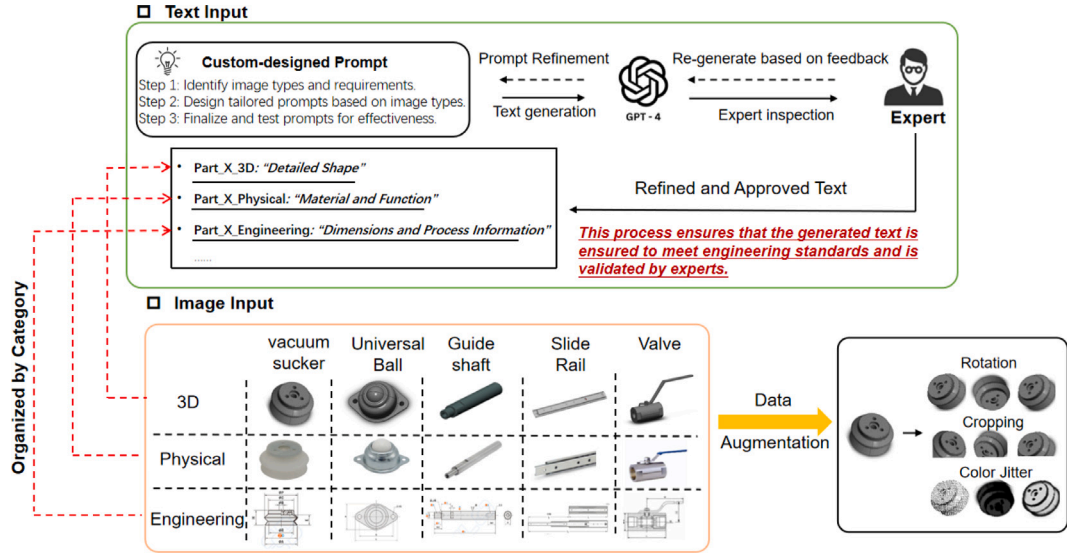


Fig. 4. Data collection and processing workflow for IPMD.

Table 8
Text data composition and instruction types in IPMD.

Type	Slide rails	Guide shaft	Universal ball	Vacuum sucker	Valves	Total
Description	45	42	27	18	9	141
Selection	6	5	4	4	3	22
Weight	4	3	2	4	2	15
Alignment	8	6	5	4	2	25
Format	1	1	1	1	1	5
Operation	1	1	1	1	1	5
Process	1	1	1	1	1	5
Environment	1	1	1	1	1	5
Total	67	60	42	34	20	223

Table 9
Generated sample pairs for model training.

Type	Positive	Negative	Total
Slide rails	750	2250	3000
Guide shaft	421	1263	1684
Universal ball	126	378	504
Vacuum sucker	771	2313	3084
Valves	152	456	608
Total	2220	6660	8880

the model. This process is repeated five times, ensuring each fold is utilized once as the validation set.

4.1.3. Cross-validation

To address the issue of model overfitting caused by dataset imbalance, this study employs a 5-fold cross-validation method. Specifically, the dataset is partitioned into five non-overlapping subsets. In each iteration, one subset is utilized as the validation set, while the remaining four subsets are used for training. This process is repeated five times. This method reduces the impact of random data partitioning to a certain extent and more significantly mitigates the risk of overfitting.

Experiments were conducted on a server running Ubuntu 22.04.4 LTS, with a 6.5.0-44-generic kernel version. Computing tasks were accelerated by NVIDIA A100 GPUs, utilizing PyTorch 2.0.1 and Python 3.8. Text processing was handled using the Meta_LLaMA_3_8B model from the Hugging Face Transformers library. The training configuration included an initial learning rate of $1e-4$, AdamW optimizer, cosine annealing scheduler to decay the learning rate from $1e-4$ to $1e-6$, batch size of 32, and 100 training epochs with the best model saved after each epoch. Image preprocessing involved resizing to 224×224 resolution with standardization (mean 0.485, 0.456, 0.406; standard

deviation 0.229, 0.224, 0.225), and data augmentation techniques like random cropping, flipping, and color jittering. To prevent overfitting, L2 regularization (weight decay $1e-2$) and dropout techniques were used, and 5-fold cross-validation was applied to enhance model generalization performance.

4.1.4. Evaluation metrics

Four evaluation metrics were used to assess the performance of the multimodal fine-grained reasoning approach: Accuracy, Precision, Recall and F1-Score. The definitions of these metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

Where TP , TN , FP , and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively. Precision measures the accuracy of positive predictions, Recall assesses the rate of correct positive recognitions, and F1-Score, as the harmonic mean of Precision and Recall, is suitable for evaluating imbalanced data. The evaluation process involves calculating each metric per category and then summarizing the overall performance.

4.2. Model evaluation

4.2.1. Experimental design idea

When researching the application of LLMs in mechanical component recognition, to comprehensively evaluate their performance, two

Table 10
Multi-model performance comparison.

Model name	LU	TG	KBQA	SS
LLaMa	68.5%	18.2	69.8%	0.70
LLaMa2	75.3%	23.5	75.6%	0.78
GPT3.5	80.2%	26.8	81.0%	0.82
LLaMa3	85.6%	29.4	86.3%	0.88

Table 11
LLaMa3's recognition performance for different mechanical components.

Components	Prompt	Recognition Acc
Guide shaft	N	25%
	Y	65%
Vacuum sucker	N	20%
	T	50%
Universal ball	N	15%
	Y	45%
Slide rail	N	30%
	Y	60%
Valve	N	22%
	Y	52%

key experiments were carried out: the multi-model performance comparison experiment and the LLM mechanical semantic understanding experiment. In the multi-model performance comparison experiment, based on the representativeness, open-source nature of the models, and their extensive application in the field of natural language processing, LLaMa, LLaMa2, GPT-3.5, and LLaMa3 were selected for comparison. Tests were conducted on tasks such as language understanding (LU), text generation (TG), Knowledge-Based Question Answering (KBQA), and semantic similarity (SS). The results are shown in Table 10.

The LLM mechanical semantic understanding experiment focused on different mechanical components, under two scenarios: without prompts and with prompts. The experimental results, as shown in Table 11, indicate that in the scenario without prompts, the accuracy of LLaMa3 in identifying guide shafts and extracting key features is relatively low, failing to meet the requirements for fine-grained recognition of mechanical components. In the scenario with prompts, although the accuracy has improved to some extent, there are still issues with inaccurate identification of some key features. The comparison between the two scenarios highlights the inherent limitations of LLaMa3 in fine-grained recognition of mechanical single-modal images.

In summary, LLaMa3 has strong general-purpose performance, but it has limitations in fine-grained recognition when dealing with mechanical single-modal images. Therefore, comparative and ablation experiments were designed. The comparative experiment compares the results of LLaMa3's independent recognition with those of a multi-modal model combining CNN and UFFM to verify the effect of multi-modal fusion. The ablation experiment removes components such as UFFM to observe changes in model performance, clarify the role of each component in multi-modal recognition, and thus optimize the model performance.

4.2.2. Comparison experiments

The comparative analysis evaluates the performance of the proposed multimodal fine-grained reasoning approach against various baselines and enhanced baselines under consistent experimental conditions. Baseline models include VisualBERT, CLIP, BLIP, and BLIP2, which models have shown excellent performance in many multimodal-related tasks. The IPMD dataset was used in the experiment. The dataset now covers five types of industrial components: vacuum suckers, guide shafts, universal balls, slides and valves. The metrics used include: accuracy, precision, recall and F1-score. The results of the comparative experiment are shown in Table 12. The proposed model is superior to the baseline model in most aspects, demonstrating its effectiveness in handling complex industrial part identification tasks. Although CLIP,

Table 12
Multi-model performance comparison.

Model	Acc	Prc	Rec	F1
VisualBERT	0.70	0.68	0.66	0.67
CLIP	0.72	0.70	0.68	0.69
BLIP	0.75	0.73	0.71	0.72
BLIP2	0.78	0.76	0.74	0.75
Our model	0.83	0.81	0.85	0.87

Table 13
Model performance comparison.

Model	Acc	Prc	Rec	F1
Full	0.87	0.89	0.83	0.84
W/O UFFM	0.74	0.76	0.73	0.71
RNS	0.79	0.75	0.74	0.8
Text	0.8	0.81	0.78	0.77
Image	0.65	0.62	0.61	0.67

BLIP, BLIP2 and VisualBERT have strong multimodal data processing capabilities, they are less adaptable to the unique characteristics and requirements of industrial part identification than our model.

Overall, the results validate the proposed model's superiority in multimodal fine-grained reasoning, with pronounced advantages in cross-modal feature alignment and consistency over baseline models. This performance is particularly advantageous in complex industrial environments that necessitate high semantic precision and resilience against noisy or incomplete inputs, underscoring the practical applicability of the proposed approach in such contexts.

4.2.3. Ablation experiment

The ablation study evaluated the contributions of each module in the model towards multimodal fusion and fine-grained recognition. The specific settings included:

- (1) Removing the UFFM module and replacing it with simple feature concatenation to assess its role in multimodal fusion;
- (2) Substituting the positive and negative sample generation mechanism with Random Negative Sample (RNS) to examine its impact on fine-grained classification;
- (3) training with only image or text input to analyze the model's dependency on each modality.

Table 13 illustrates the performance impact of ablative modifications on the proposed model, underscoring the critical contributions of specific components within the multimodal fine-grained reasoning framework. The removal of the UFFM led to marked reductions in F1-Score from 0.84 to 0.71 and Accuracy from 0.87 to 0.74, which demonstrates UFFM's pivotal role in enhancing multimodal feature interaction and ensuring coherent integration of textual and visual information. This decline reflects the limitations of simpler fusion approaches that fail to adequately capture the nuanced interdependencies between modalities.

Similarly, substituting the dedicated positive and negative sample generation mechanism with random negative sampling caused a substantial decrease in Precision from 0.89 to 0.76 and Recall from 0.83 to 0.75. This reduction highlights the specialized sample generation mechanism's essential function in refining the model's sensitivity to fine-grained differences between similar classes. The targeted generation of high-quality negative samples effectively sharpens the model's discriminatory capabilities.

Moreover, the single-modality configuration, especially the image-only modality, shows the lowest overall performance, with the F1-score and accuracy dropping to 0.67 and 0.65, respectively. This stark contrast emphasizes the inherent limitations of single-modality approaches, which lack the comprehensive perspective afforded by multi-modal fusion. The text-only modality, although performing better than the image-only setup, still fell short in capturing complex interactions, affirming the necessity of cross-modal information synthesis to achieve

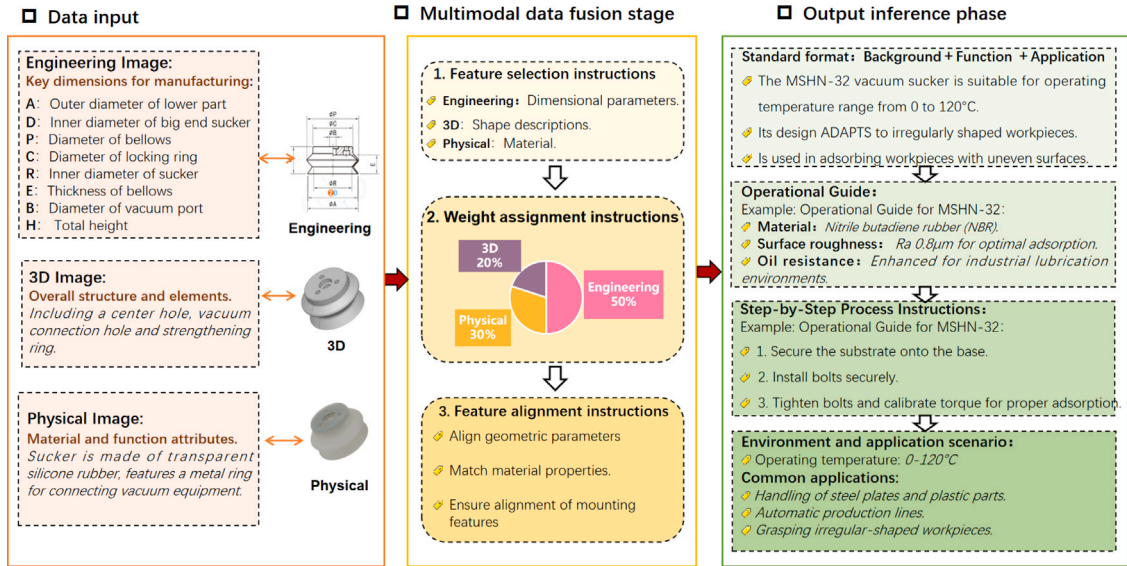


Fig. 5. Query process for MSHN-32 heavy-duty vacuum sucker.

Multimodal Parts Identification and Recommendation System

This system allows users to upload an image or enter a query description to find matching industrial parts.

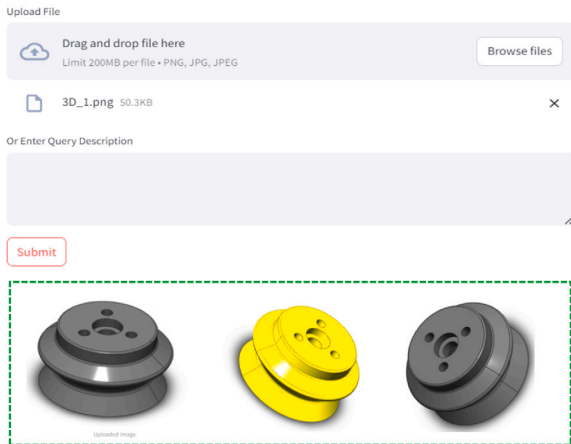


Fig. 6. Image input scenario for multimodal query system.

superior classification performance.

In conclusion, Table 13 validates the indispensable role of both the UFFM and the specialized sample generation mechanisms in enhancing model robustness and fine-grained recognition accuracy, particularly in complex industrial scenarios requiring multimodal data integration.

5. Case study

5.1. Case background

This case study focuses on the application of a multimodal fine-grained reasoning model within a complex industrial setting, specifically demonstrating the model's capability through the interactive visualization system using the MSHN-32 heavy-duty bellows-type Vacuum sucker as an example. Fig. 5 illustrates the entire process flow, beginning with data input, followed by multimodal data fusion, and culminating in output inference. The diagram emphasizes how each

Multimodal Parts Identification and Recommendation System

Upload an image or enter a query description to find matching industrial parts.



Fig. 7. Text input scenario for multimodal query system.

data type—engineering drawings, 3D images, and physical images—is utilized and integrated at various stages to generate actionable guidance.

5.2. Method application and results analysis

The system offers two distinct query methods: image input and text input, allowing users to engage with the system through different modalities. Fig. 5 provides a detailed overview of how the multimodal data fusion stage operates. For image inputs, the engineering drawings capture critical manufacturing dimensions, such as outer diameters, thicknesses, and other specifications necessary for ensuring part compatibility. The 3D images provide structural details, reinforcing the geometric understanding required for precise recognition. Physical images contribute material and surface property information, vital for validating the mechanical suitability of the part.

As shown in Fig. 6, when an image of the suction cup is uploaded, the system employs a CNN to extract visual features. These features are then matched against a database of part characteristics, calculating similarity scores that inform the identification process. The diagram emphasizes the importance of each data input type: engineering drawings deliver exact dimensional data, 3D images offer structural insights, and physical images confirm material aspects.

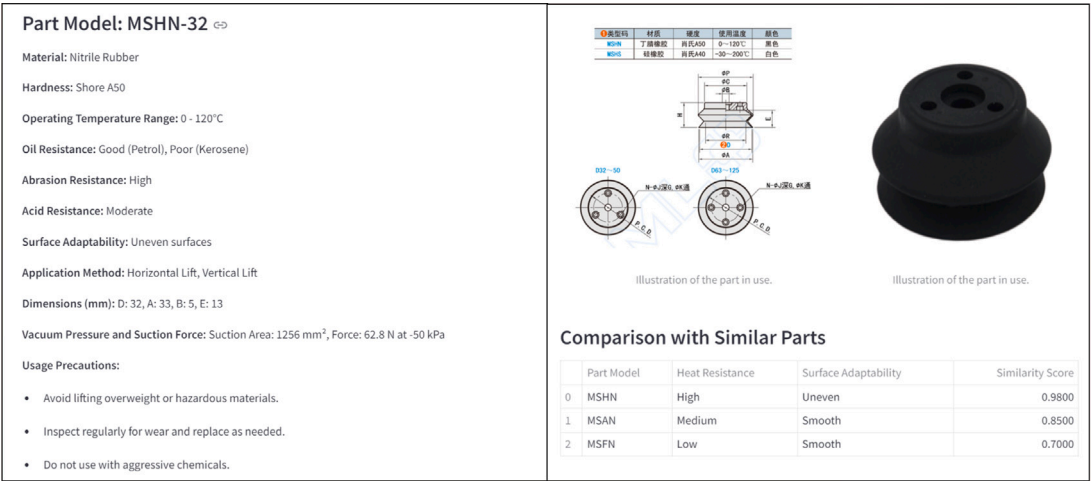


Fig. 8. System recommendations based on image input analysis.

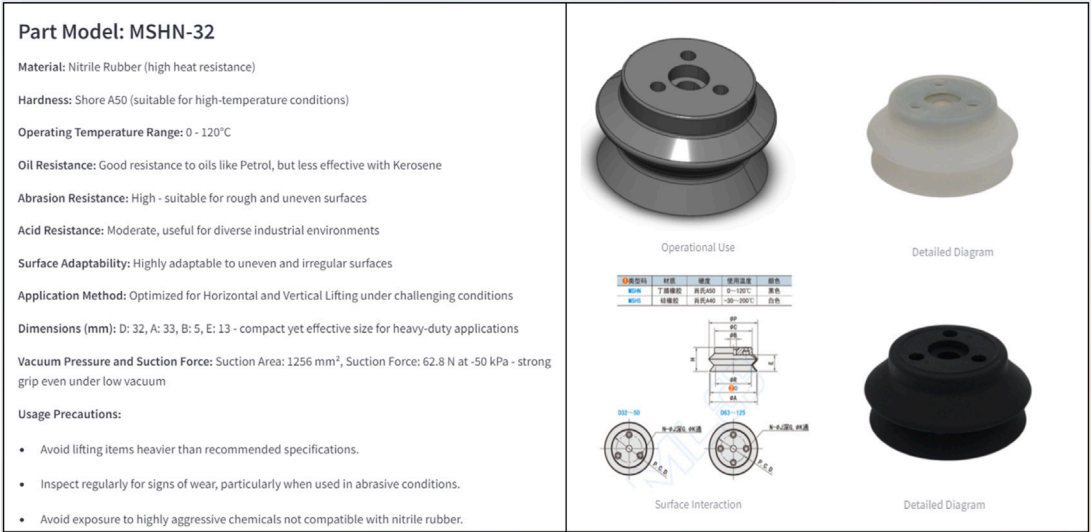


Fig. 9. System recommendations based on text input analysis.

Fig. 7 depicts the scenario where the user inputs textual descriptions such as "Vacuum sucker for high-temperature and uneven surfaces". The system leverages a large language model to parse these descriptions, extract textual features, and integrate them with the visual features extracted from images. This fusion enables a comprehensive analysis that accurately identifies the MSHN-32 suction cup, demonstrating the collaborative interaction of multimodal data.

Query results are shown in Figs. 8 and 9, where the system recommends part models that match the conditions specified in the image and text queries, respectively. In the image input scenario, shown in Fig. 8, the system performs deep feature analysis on the MSHN-32 suction cup image and outputs detailed part information, including material, heat resistance, corrosion resistance, and surface adaptability, along with comparisons to similar parts. Additionally, other image types, such as engineering drawings and real-world images, are provided. Engineering drawings offer precise dimensions and structural details that aid in understanding design specifications, while real-world images show the actual appearance and application scenarios of the suction cup, further enhancing the understanding of its characteristics.

In the text input query, as shown in Fig. 9, the system accurately identifies and recommends the MSHN-32 suction cup, which exhibits excellent heat resistance and adaptability to uneven surfaces, meeting the user's requirements. The system also provides additional

detailed information about the suction cup, including material properties, corrosion resistance, usage methods, and related precautions, fully explaining the recommendation. Beyond fulfilling the initial query, the system expands the content by returning three types of images: 3D models, engineering drawings, and real-world images. The 3D models display the part's three-dimensional structure, the engineering drawings detail the dimensions and design specifications, and the real-world images visually present the suction cup in practical applications.

5.3. System performance evaluation and improvement directions

The case study demonstrates that the multimodal query system effectively improves the accuracy of complex part recognition, especially for the selection of the MSHN-32 heavy-duty Vacuum sucker. By integrating the UFFM, the system achieves deep multimodal information fusion, significantly enhancing the collaborative understanding of visual and textual features. Experimental results confirm this, showing that the proposed method outperforms baseline models in key metrics such as accuracy, precision, recall, and F1 score. Notably, the system's overall accuracy reached 87%, exceeding that of traditional unimodal methods by more than 20%, highlighting its superior fine-grained recognition capability.

6. Conclusion

This research presents a multimodal fine-grained reasoning framework designed to address the complex challenges of recognizing and semantically interpreting mechanical components in industrial product design systems. By leveraging the combined strengths of CNNs for visual feature extraction, LLaMA3 for textual semantic processing, and UFFM for cross-modal feature synthesis, this approach achieves significant advancements over traditional unimodal methods, particularly in high-precision industrial applications. The following contributions highlight the key achievements of this study:

The proposed innovative mechanical part semantic understanding approach is verified to be effective on cross-modal alignment, feature consistency, and comprehensive semantic reasoning. By integrating CNNs, LLaMA3 and UFFM, the proposed framework outperforms baseline models by up to 20% in accuracy on semantic understanding task, especially in scenarios requiring nuanced comprehension and high precision. Meanwhile, this approach achieves high cross-modal alignment and feature consistency, validating its effectiveness in complex industrial contexts where precise part recognition is essential. These gains are noteworthy in environments demanding exacting detail.

A UFFM was proposed for fine-grained feature capture and cross-modal interactions. It is validated through ablation experiments. The experiment results underscore the importance of UFFM and the positive-negative sample generation mechanism, both of which drive robust feature fusion and fine-grained classification, enhancing the model's capability to handle multimodal data.

A dataset IPMD and an interface are built to showcase the proposed approach and for other researchers' reference. The IPMD contains engineering diagrams, 3D model screenshots, and textual descriptions of these figures, which could be used for multi-modal product design. The visual interface improves the interpretability of the model's reasoning process, fostering user trust and supporting deployment in industrial applications where transparency is vital.

However, the static feature weighting strategy currently employed limits the model's adaptability across tasks, and the restricted scope and diversity of the training dataset present challenges to generalizability. Additionally, the model shows sensitivity to variations in geometry and material, indicating that further improvements in multimodal feature integration are necessary. Therefore, our future research will focus on developing adaptive feature weighting mechanisms. By introducing context-aware or task-aware approaches, the task contexts could be fully understood at the semantic level, and the features could be dynamically weighted according to the semantic information of different contexts. Besides, our dataset will be expanded to include a brother variety of components and industrial settings, thereby enhancing the model's robustness and applicability in diverse industrial scenarios.

CRedit authorship contribution statement

Xinxin Liang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Zuoxu Wang:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Jihong Liu:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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